



# EXPERIMENT TO DETERMINE THE MAGNITUDE OF THE MAGNETIC PERMEABILITY OF FREE SPACE

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## AIM

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To determine the magnitude of the magnetic permeability of free space.



## OBJECTIVES

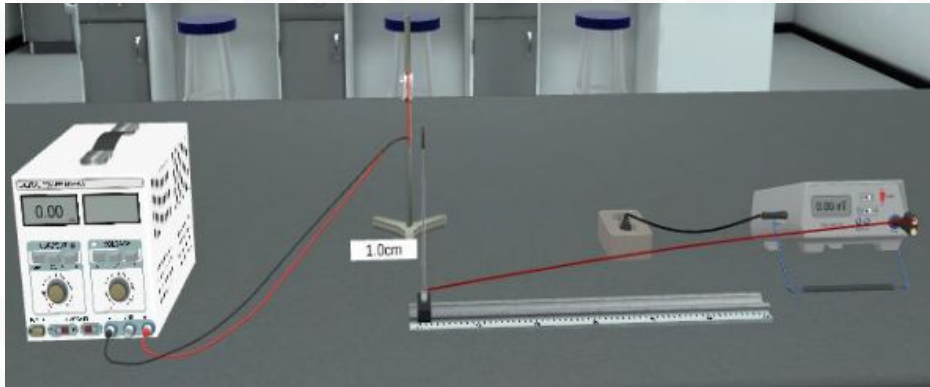
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1. Measure and map the magnetic flux density along the axis of a circular loop.
2. Identify and analyse the key factors that affect magnetic flux density in the loop.
3. Investigate and understand the relationship between the intensity of the current passing through the circular loop and the resulting magnetic field strength.
4. Utilize experimental data to calculate the magnetic permeability of free space ( $\mu_0$ ), providing a quantitative understanding of the magnetic field's behaviour.



## APPARATUS

- A circular loop of radius  $R$ .
- DC power supply.
- Ammeter.
- Movable magnetic probe.
- Tesla meter.





## THEORY

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- Hans Christian Oersted in 1820 discovered, that when a compass is placed near a current–carrying conductor, its needle deflects by an angle that is proportional to the magnitude of the current.
- The degree of deflection is directly proportional to the current's magnitude.

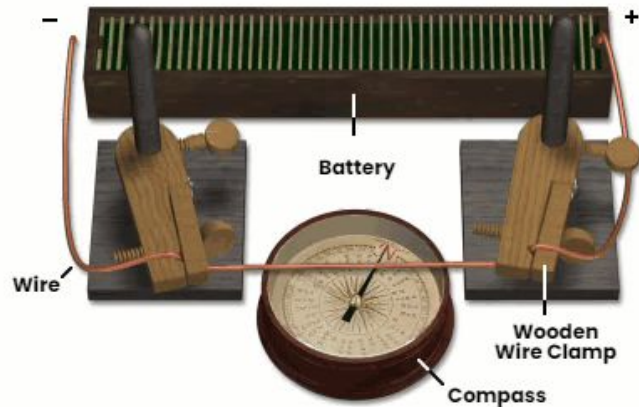


**Hans Christian Ørsted**  
(1777 – 1851)



## THEORY

This ground-breaking observation in electromagnetism laid the foundation for understanding the relationship between electricity and magnetism.



Flip Battery

☐ Off



**Hans Christian Ørsted**  
(1777 – 1851)





## THEORY

The magnetic flux density,  $B(z)$  due by a circular loop carrying current  $I$  and consisting of one turn, radius  $R$ , at a point on the axis of the loop located at distance  $z$  from its centre is given by:

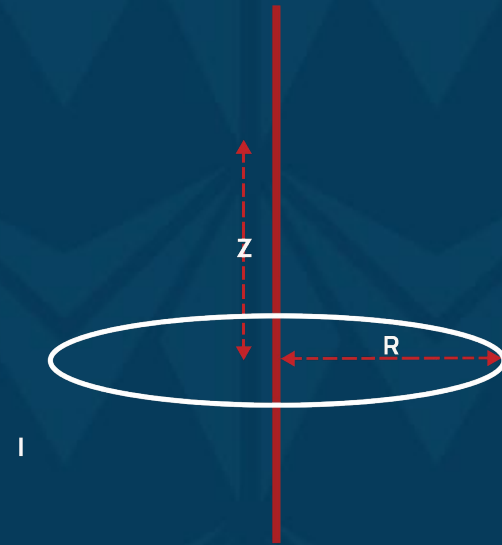
$$B(z) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$$

$B(z)$ : Magnetic flux density at a point on the axis of the loop (distance  $z$  from the center).

$\mu_0$  (Permeability of Free Space):  $4\pi \times 10^{-7} \text{ T.m/A}$ .

$R$ : Radius of the circular loop.

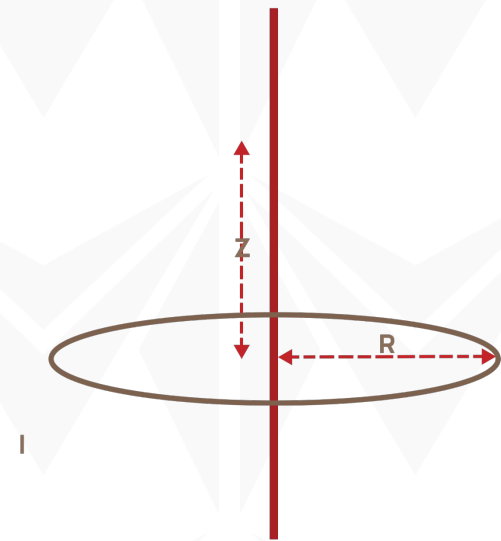
$I$ : Current flowing through the loop.





## THEORY

- The magnetic flux density,  $B$  has its maximum value at the centre of the loop  $z = 0$  and decreases as we move away from it along the axis of the loop.
- Permeability is a measure of how easily a material can be magnetized



$$B(z) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$$

$$y = m x$$





## MID-LESSON QUESTIONS

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Question 1:

What did Hans Christian Oersted discover when he placed a compass near a current-carrying conductor?

Question 2:

In the formula for magnetic flux density  $B(z)$  due to a circular loop carrying current  $I$ , what does the symbol  $\mu_0$  represent, and why is it significant in our experiment?



## MID-LESSON ANSWERS

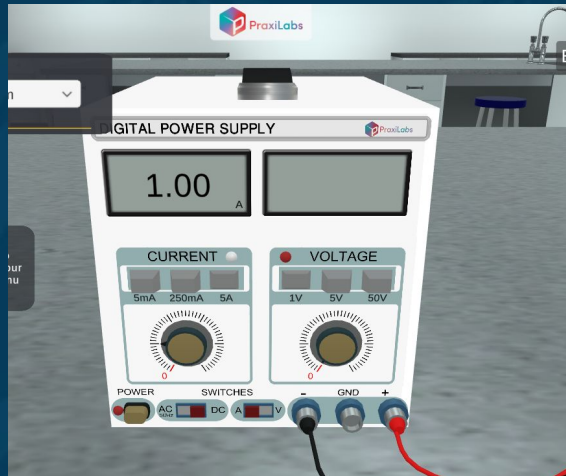
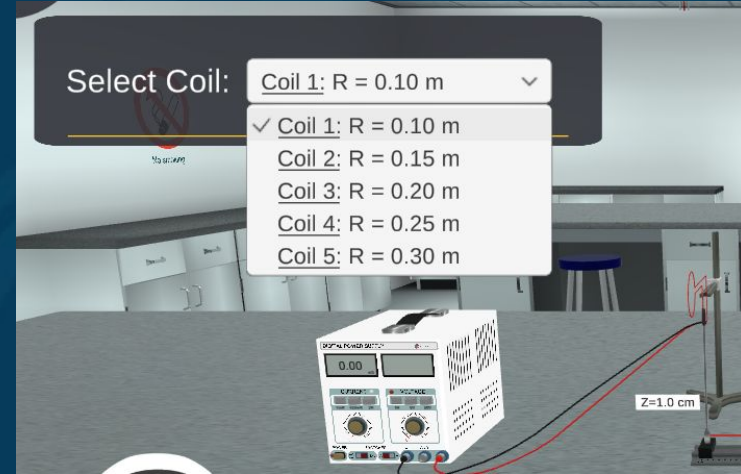
Answer 1: Hans Christian Oersted discovered that when he placed a compass near a current-carrying conductor, the compass needle deflected. The degree of deflection was found to be proportional to the magnitude of the current.

Answer 2: In the formula for magnetic flux density  $B(z)$  due to a circular loop carrying current  $I$ , the symbol  $\mu_0$  represents the permeability of free space. It is significant because it quantifies the relationship between magnetic field strength and the current in the loop, as well as the geometric properties of the loop. It serves as a fundamental constant in electromagnetism, allowing us to calculate the magnetic field in a given situation.



## PROCEDURE

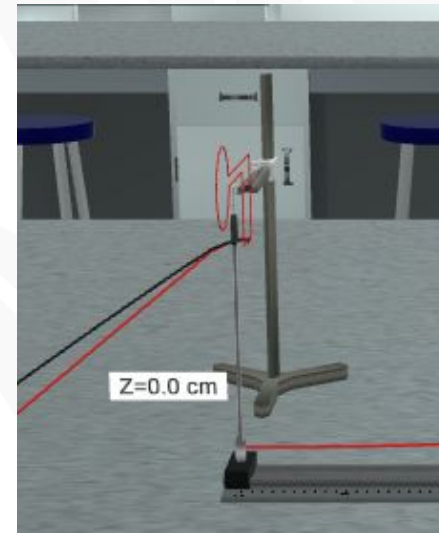
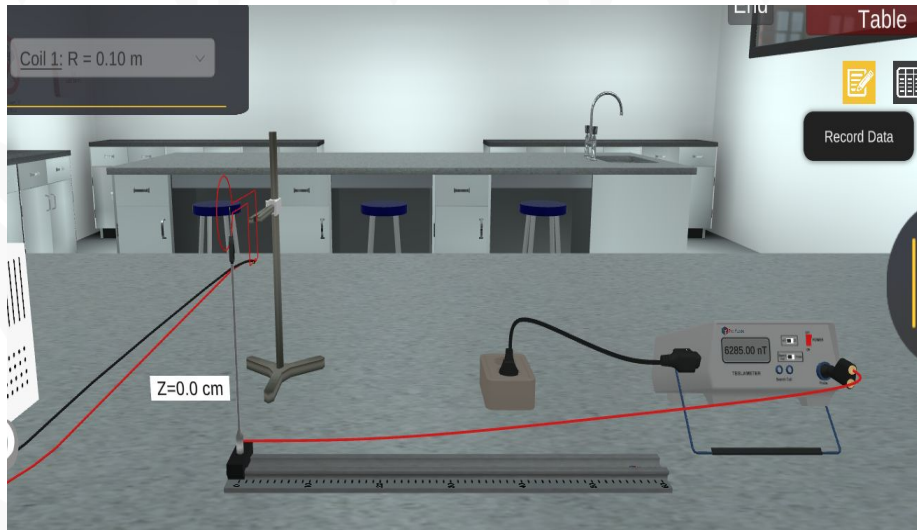
1. Select the default coil 1:  $R = 0.10 \text{ m}$
2. Adjust the current to 1 A (Other current values can be explored) on the digital power supply.





## PROCEDURE

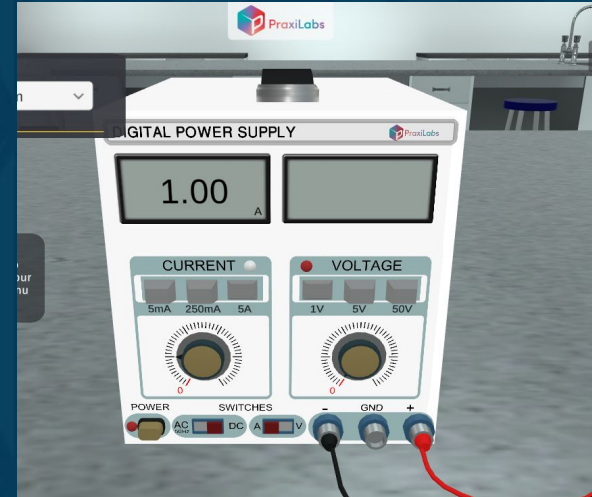
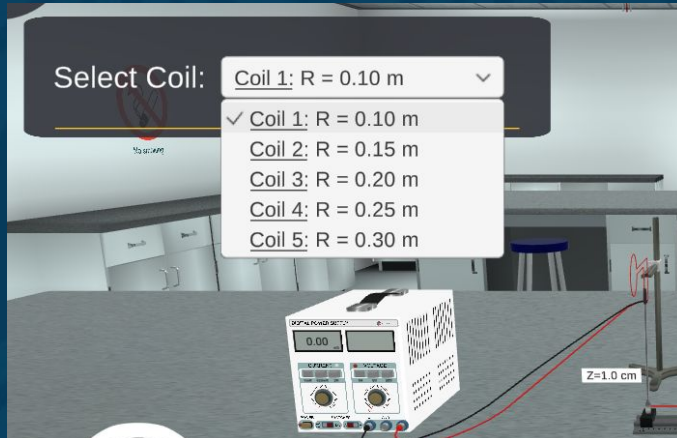
3. Move the magnetic probe so that it lies at the centre of the coil ( $z = 0$ ).
4. Click the Record button to store the result that appears on the Tesla meter.





### PROCEDURE

5. Increase the distance  $z$  in step of 2 cm (2, 4, 6 etc.) then record the new value of  $B$  in each case (up to 20 times in all)
6. Repeat the above steps for different coils (Coil 2, Coil 3 etc.).





## PROCEDURE

- Once the experiment has been completed, an Excel sheet containing the recorded results will be downloaded.

### Magnetic field of circular loop current

|      |     |  |        |         |
|------|-----|--|--------|---------|
| R(m) | 0.1 |  |        |         |
| I(A) | 1   |  |        |         |
|      |     |  |        |         |
|      |     |  | Z (cm) | B (nT)  |
|      |     |  | 0      | 6285    |
|      |     |  | 2      | 5928.99 |
|      |     |  | 4      | 5039.95 |
|      |     |  | 6      | 3950.37 |
|      |     |  | 8      | 3000.93 |
|      |     |  | 10     | 2220.22 |
|      |     |  | 26     | 289.73  |
|      |     |  | 28     | 238.54  |
|      |     |  | 30     | 199.35  |
|      |     |  | 32     | 166.43  |
|      |     |  | 34     | 140.82  |
|      |     |  | 36     | 120.59  |
|      |     |  | 38     | 103.35  |



## DATA ANALYSIS

1. Using the downloaded Excel data, calculate the factor,

$$\frac{R^2}{(R^2 + z^2)^{3/2}}$$

2. Plot the magnetic field, B vs. the computed factor in 1 above.
3. Calculate the slope of the graph.
4. Calculate the permeability of free space from the formula.

$$y = m x$$
$$B(z) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$$

$$B(z) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$$

|      |     |       |                                 |
|------|-----|-------|---------------------------------|
| R(m) | 0.1 |       |                                 |
| I(A) | 1   |       |                                 |
|      |     | Z(cm) | $\frac{R^2}{(R^2 + z^2)^{3/2}}$ |
|      |     | 0     | 6285                            |
|      |     | 2     | 5928.99                         |
|      |     | 4     | 5039.95                         |
|      |     | 6     | 3950.37                         |
|      |     | 8     | 3000.93                         |
|      |     | 10    | 2220.22                         |
|      |     | 26    | 289.73                          |





## WALK THROUGH VIDEO







## FURTHER READING

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## **FURTHER READING**

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<http://hyperphysics.phy-astr.gsu.edu>

<http://www.physicstutorials.org>



# THANK YOU